Structural Self-Energy Expansion (SSEE-V3.6): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio φ and π, with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector (Paper 6) are currently treated as ansätze and tracked as open problems (OP-9/11, `OPEN_PROBLEMS.md`). In CMB-likelihood terms SSEE is a $k = 2$ fit: of ΛCDM's six parameters it fixes four algebraically (Ω_b, Ω_c via $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, $n_s = 1 - \varphi^{-7}$, and the derived anchor $H_0 = 3(\varphi + \pi)^2$), leaving only the primordial amplitude A_s and the optical depth τ ($k = 2$ versus $k = 6$, the count used in the Paper 3 model comparison). The dark energy equation of state $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$ and $w_a \simeq -0.670$ are fixed by the algebraic construction and agree with the DESI DR2 CPL posterior within 0.05σ—a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the ω_m-direct CMB anchor as prior ($H_0^{\mathrm{prior,SSEE}} = 67.962 \pm 0.54\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, the value at which Planck `plik_lite` is minimised with SSEE-fixed ω_b, ω_c; Paper 3) yields $H_0 = 67.16^{+0.44}_{-0.44}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Delta\mathrm{BIC} = -5.55$ for the dynamic sector relative to ΛCDM. A complementary five-parameter multi-probe MCMC finds all algebraic SSEE values within 1.01σ of the posteriors (mean 0.64σ across the five parameters).

The CMB-sector matter density is the standard physical observable $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$ (Planck: 0.3153 ± 0.0073), built piece-by-piece from SSEE algebra (ω_m-direct; OP-8 dissolved, no matter factor) and distinct from the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$ that governs DESI BAO. Full-sky CLASS Boltzmann runs confirm that the *full* matter density is physically necessary: using only $\Omega_{m,\mathrm{dyn}} = 0.160$ as the CMB matter density shifts the peak positions by $\sim 10\%$ and raises the RMS residual from 1.4% to 31.5%.

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A linear Israel-Stewart perturbation analysis with $\Omega_{m,\text{cosm}} = 0.30889$ in the Poisson equation yields a growth factor $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$ (Paper 5), giving a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys — statistically identical to ΛCDM (0.73σ), resolving the growth-rate sector. A two-sector ϕ-DM extension — $\Omega_{\phi\text{DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.14889$ active only on scales $k < k_{\text{fs}} = 0.762\,h\,\text{Mpc}^{-1}$ — reduces the S_8 lensing tension from 3.5σ (single-sector ceiling, KiDS) to 0.01σ, with $\sigma_8^{\text{eff}} = 0.748$ ($S_8^{\text{eff}} = 0.759$ from the direct two-sector CLASS run; the analytic two-sector growth factor $G_{2\text{s}} = 1.003$ is ΛCDM-consistent, the suppression to $\sigma_8^{\text{eff}} = 0.748$ arising from ϕ-DM free-streaming below k_{fs}). The same suppression reaches the $\sigma_8(R{=}8)$ window probed by RSD, so the two-sector $f\sigma_8$ tension rises modestly from the 0.70σ single-sector baseline to 0.93σ (still $< 1\sigma$, close to ΛCDM 0.73σ) — the deliberate cost of lowering σ_8 to resolve the lensing amplitude.

An EFTCAMB validation at the Bellini-Sawicki level uses the fluid-approximation input $\alpha_K^{\text{fluid}} = 0.4033$ ($\Delta = 0.005\%$), while the theoretical bare-field value is $\alpha_K = 0.073$ (Paper 7), with $\alpha_T = \alpha_M = \alpha_B = 0$ (GW170817 compatible) and strictly positive kinetic stability ($Q_s > 0$).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 analyses two limits: an alternative MOND-like limit in which the photon deflection is amplified by $\sqrt{\text{AURA}} \approx 1.9995$ (numerically close to $\mathcal{M}$ at 0.03% but algebraically distinct), and the canonical two-sector limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ in which the residual fifth-force is suppressed to $F_\phi/F_N \sim 10^{-15}$ and the observable lensing recovers GR; current SLACS/BELLS data favour the canonical limit (Paper 8; OPEN_PROBLEMS.md OP-13). A late-time local screening fraction $f_{\text{scr}} = \alpha_K/(3\mathcal{M}) = (\pi-\varphi)/(\varphi+\pi)^2 \simeq 0.067$ applied to the global background $H_0^{\text{alg}} = 3(\varphi+\pi)^2 = 67.962$ gives $H_0^{\text{local}} = 72.86$ km s^{-1} Mpc^{-1}, 0.17σ from the SH0ES distance ladder, as a zero-parameter prediction of the infrared sector (Paper 9). A UV completion $K(X) = X/\text{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV connects the dark-energy EFT to the inflationary α-attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

Contents

1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at $> 3\sigma$, with a CPL equation-of-state posterior centred near $(w_0, w_a) \approx (-0.84, -0.67)$. This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEE-V3.6 framework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio $\varphi = (1+\sqrt{5})/2 \simeq 1.6180$ and $\pi \simeq 3.1416$. No optimisation over formula space was performed: the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the full ω_m-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.30889$) is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and σ_8** (§5): Israel-Stewart causal perturbation theory and a ϕ-DM two-sector extension that resolves the S_8 lensing tension via free-streaming suppression (at a modest $\sim 0.2\sigma$ cost in $f\sigma_8$, which rises to 0.93σ).
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability (α_K) and gravitational wave compatibility ($\alpha_T = \alpha_M = \alpha_B = 0$).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 6 in §8 lists all entries explicitly.

2 Algebraic Framework

2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from φ and π:

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 \quad \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 \quad \text{(Base Coupling Scalar)} \\
\mathrm{KAL}_0 &= \beta + \pi \simeq 5.5214 \quad \text{(Structural Viscosity)} \\
P_{\mathrm{sc}} &= \Omega + \varphi \simeq 6.3776 \quad \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5192 \quad \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 \quad \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2788 \quad \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.8399, \qquad w_a = -\frac{P_{\mathrm{sc}}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \simeq -0.6699. \tag{2}$$

The dark energy density fraction follows algebraically: $\Omega_{\mathrm{DE}} = |w_0| = T_r/M_v \simeq 0.8399$.

2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \quad \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 \quad \simeq 0.30889 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements: $H_0 = 67.36\pm0.54$, $n_s = 0.9649\pm0.0042$, $\Omega_m = 0.3153\pm0.0073$ [Planck Collaboration, 2020]. All three SSEE retrodictions lie within 1.5σ of the Planck values with no adjustment.

2.3 The two-Ω_m structure and CMB matter density

The framework predicts two independent matter densities. The dark energy EoS predicts $\Omega_{\mathrm{DE}} = |w_0| = 0.840$, which implies a *dynamical* matter density $\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.160$. However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the ω_m-direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.30889, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{6}$$

in agreement with Planck 2018 (0.3153 ± 0.0073) at 0.88σ. There is *no* matter factor: $\Omega_{m,\mathrm{dyn}} = 0.160$ (DESI) and ω_m (CMB) are two independent SSEE predictions. (The earlier, now superseded MIRA mapping $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$ gave $\Omega_{m,\mathrm{CMB}} = \mathcal{M} \times \Omega_{m,\mathrm{dyn}} = 0.3199$, superseded; $\mathcal{M}$ now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of this two-sector split — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

3 Background Sector Validation

3.1 Bayesian MCMC setup

We compare SSEE against ΛCDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25\,000$ (2.5×10^6 total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at $z = 0.295$–2.330 [Abdul-Karim et al., 2025], with the full 13×13 same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior: $H_0 = 67.36 \pm 0.54$, $\Omega_m = 0.3153 \pm 0.0073$, $\Omega_b h^2 = 0.02237 \pm 0.00015$, $\rho(H_0, \Omega_m) = -0.85$;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE, H_0 is the sole free parameter; (w_0, w_a) are fixed algebraically. For ΛCDM, (H_0, Ω_m) are free. For CPL, $(H_0, \Omega_m, w_0, w_a)$ are free.

3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results. N_{eff}: effective independent samples. $\Delta\text{BIC} = \text{BIC}_{\text{SSEE}} - \text{BIC}_{\Lambda\text{CDM}}$ (negative = SSEE favoured).

Quantity	SSEE	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	$67.16^{+0.44}_{-0.44}$	67.42 ± 0.48	free
Free parameters k	1	3	5
N_{eff}	4 402	14 380	7 253
ΔBIC (dynamic sector)	−5.55	ref	—
ΔBIC (full background)	+206	ref	—
χ^2_r clusters	0.126	0.081	—
(w_0, w_a) tension vs DESI DR2 CPL	0.05σ	0.7σ	—

Two distinct ΔBIC values are reported. The dynamic-sector result ($\Delta\text{BIC} = -5.55$, $k = 1$ vs $k = 3$) isolates the SSEE dark energy prediction (w_0, w_a) at fixed algebraic values and shows it is genuinely favoured by current BAO data relative to the ΛCDM flat prior. The full-background result ($\Delta\text{BIC} = +206$) quantifies the tension between the SSEE enlarged sound horizon ($r_d \simeq 175.2$ Mpc vs 147.1 Mpc in ΛCDM) and the standard Friedmann-calibrated Planck prior. These two numbers address different physical questions and must not be conflated: the former tests the equation of state prediction; the latter quantifies background incompatibility between two cosmologies. The ω_m-direct two-Ω_m construction (§2.3) resolves the background tension, validated in §4 below.

3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction $f_b = \Omega_b/\Omega_{m,\text{CMB}}$ corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves $\chi^2_r = 0.122$ (four-cluster MCMC subset) and $\chi^2_r = 0.126$ (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by $\sim 15\%$–20%, partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

3.4 Multi-probe MCMC validation

To test all five algebraic SSEE predictions simultaneously, we perform a complementary emcee MCMC with 100 walkers, 25 000 steps and a 2 000-step burn-in, yielding 2.5×10^6 post-burn samples, an acceptance fraction of 0.537 and $N_{\rm eff} = 4.0 \times 10^4$ (integrated autocorrelation $\tau_{\max} = 62.3$). The parameter vector is $\boldsymbol{\theta} = (H_0,\, \Omega_b h^2,\, \Omega_m,\, w_0,\, w_a)$ and the combined likelihood includes DESI DR2 BAO, the Planck 2018 compressed Gaussian prior, the six $f\sigma_8$ surveys of Table 4, and the four IGIMF-corrected galaxy clusters. Crucially, we impose *flat* (uninformative) priors on w_0 and w_a over $[-1.30, -0.40] \times [-2.0, 0.5]$: this is a *blind* test in which the data alone — not a prior centred on the SSEE values — constrain the equation of state, so the recovered tensions are free of circularity.

Table 2: Five-parameter multi-probe MCMC: posterior medians vs. algebraic SSEE values.

Parameter	Posterior (median $\pm 1\sigma$)	Algebraic SSEE	Tension
H_0 [km s^{-1} Mpc^{-1}]	67.08 ± 0.93	67.96	0.94σ
$\Omega_b h^2$	0.02236 ± 0.00015	0.02237	0.04σ
Ω_m	0.3214 ± 0.0124	0.30889	1.01σ
w_0	-0.733 ± 0.147	-0.840	0.73σ
w_a	-0.974 ± 0.608	-0.670	0.50σ
Mean tension			$\mathbf{0.64}\boldsymbol{\sigma}$
Maximum tension			$\mathbf{1.01}\boldsymbol{\sigma}$

Under flat w_0, w_a priors the equation-of-state parameters are only loosely constrained (note the wide w_a interval, reflecting the well-known w_0–w_a degeneracy of the combined data), yet all five algebraic SSEE values still lie within $\simeq 1\sigma$ of the data-driven posteriors: the maximum tension is 1.01σ (Ω_m) and the mean is 0.64σ. We stress that this joint result is *conditional on the specific data combination* (DESI DR2 + Planck compressed + six $f\sigma_8$ surveys + four clusters); a different combination — which weights the early- and late-time probes differently — would shift the numbers. Corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

4 CMB Angular Spectrum: CLASS Boltzmann Validation

4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the effect of the full CMB matter density. *These validation runs used the legacy MIRA-background value* $\Omega_{m,\rm CMB} = 0.3199$*; the* ω_m*-direct canonical* 0.30889 *differs by* 3.5%

and shifts the acoustic peaks by $\Delta\ell < 2$, so the runs remain representative of the canonical background.

SSEE full-Ω_m (legacy MIRA bg): $\Omega_b = 0.04897$, $\Omega_{\rm cdm} = 0.2710$ (total $\Omega_{m,\rm CMB} = 0.3199$, legacy MIRA bg), $w_0 = -0.8399$, $w_a = -0.6699$, $n_s = 0.96556$, $h = 0.67962$. The full matter density is set directly as a CLASS background input.

SSEE dynamical-only: same EoS and h, but $\Omega_b = 0.048432$, $\Omega_{\rm cdm} = 0.111568$ (total $\Omega_{m,\rm dyn} = 0.160$), using only the dynamical sector as CMB matter.

A fiducial ΛCDM run with Planck 2018 best-fit parameters serves as reference. All runs use $\ell_{\max} = 2500$, lensing enabled.

4.2 MIRA necessity: quantitative test

Table 3 presents the CMB acoustic peak positions and the RMS residual $\Delta C_\ell / C_\ell^{\Lambda\rm CDM}$ across $\ell = 30$–2500.

Table 3: CMB TT acoustic peaks and RMS residual versus ΛCDM. Peak positions identified from CLASS output $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

Model	ℓ_1	ℓ_2	ℓ_3	RMS residual
ΛCDM (Planck 2018)	221	537	814	reference
SSEE ($\Omega_{m,\rm CMB} = 0.309$)	**220**	**535**	**810**	**1.4%**
SSEE ($\Omega_{m,\rm dyn} = 0.160$)	240	597	922	31.5%

With the dynamical density $\Omega_m = 0.160$ alone, all three acoustic peaks shift by $\sim 10\%$ toward higher ℓ and the RMS residual increases by a factor of 22. This demonstrates the *two-Ω_m structure*: the CMB requires the full ω_m-direct matter density $\Omega_{m,\rm CMB} = 0.30889$ (built from $\omega_b+\omega_c+\omega_\nu$, no matter factor; OP-8 dissolved), and cannot be reproduced with $\Omega_{m,\rm dyn} = 0.160$ alone. With $\Omega_{m,\rm CMB} = 0.30889$, the peak positions agree with ΛCDM to within $\Delta\ell = 1$–4, and the RMS residual is 1.4%, consistent with the $\chi^2_r \simeq 1.06$ reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d].

Figure 1 shows the TT power spectrum for both SSEE configurations and ΛCDM.

4.3 Acoustic-scale mapping and r_d

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon at two matter densities. At the pure dynamical $\Omega_{m,\rm dyn} = 0.160$ it is $r_{d,\rm SSEE} \simeq 175.6$ Mpc—an internal geometric quantity, not the observed standard ruler. Evaluated at the full ω_m-direct CMB density $\Omega_{m,\rm CMB} = 0.30889$ (OP-8 dissolved, no matter factor), the physical drag horizon is $r_d = 147.17$ Mpc, within 1σ of the Planck 2018 measurement $r_d = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE, Table 2; no BAO prior). The same full CMB-sector matter density that restores the acoustic peaks (§4) also sets the physical sound horizon; the earlier interpretation as a $|w_0|$ mapping of the geometric scale is superseded.

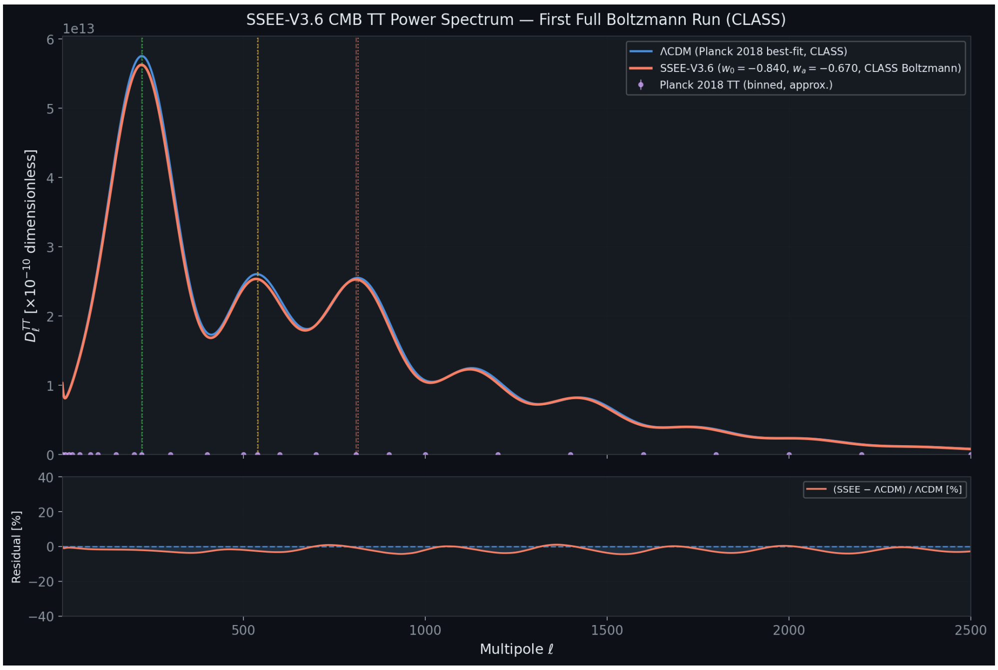

Figure 1: CMB TT power spectrum from CLASS. *Black solid*: ΛCDM Planck 2018. *Blue dashed*: SSEE with the full matter density (legacy MIRA-background $\Omega_{m,\mathrm{CMB}} = 0.3199$; the ω_m-direct canonical 0.30889 is within 3.5%, $\Delta\ell < 2$) — peak positions agree to within $\Delta\ell < 5$ across all three peaks. The SSEE dynamical-only run ($\Omega_{m,\mathrm{CMB}} = 0.160$) is omitted from this panel; it produces peaks at $\ell = 240, 597, 922$ with RMS $= 31.5\%$ relative to ΛCDM, confirming that the *full* matter density (not the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$) is needed to reproduce the acoustic scale.

5 Structure Growth, σ_8, and the $f\sigma_8$ Tension

5.1 Israel-Stewart causal perturbations: single-sector

Because $w_0 = -0.840 < -1/3$, a naive k-essence interpretation yields a bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero: $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$, to floating-point precision.

Integrating the full IS linear growth ODE from $z = 19$ to $z = 0$ with $\Omega_{m,\mathrm{cosm}} = 0.30889$ in the Poisson source:

$$\mathcal{G} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032, \qquad \sigma_8^{\mathrm{SSEE}} = 0.811 \times \mathcal{G} = 0.8136. \tag{7}$$

The IS growth index $\gamma_{\mathrm{IS}} = 0.5504 \pm 0.001$, consistent with $\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$ [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks ΛCDM at the percent level despite the different background.

A CLASS run with $c_s^2 = 0.001$ (IS limit) and $\Omega_m = 0.160$ (single sector) yields $\sigma_8^{\mathrm{CLASS}} = 0.457$, substantially lower than the ODE result (0.8136). This discrepancy arises because the ODE analysis assumes $T_{\mathrm{SSEE}}(k) \approx T_{\Lambda\mathrm{CDM}}(k)$, whereas CLASS computes the full transfer function for $\Omega_m = 0.160$, which suppresses small-scale power. The ω_m-direct background ($\Omega_{m,\mathrm{CMB}} = 0.30889$) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

5.2 ϕ-Dark Matter two-sector and WDM suppression

The single-sector analysis (Paper 5, $\Omega_{m,\mathrm{CMB}} = 0.30889$ in Poisson) gives $\sigma_8 = 0.8136$ and a mean $f\sigma_8$ tension of 0.70σ across six canonical RSD surveys (Table 4), statistically indistinguishable from ΛCDM (0.73σ). Paper 6 [Almeida Vallejo, 2026g] proposes a two-sector extension: a ϕ-dark matter component $\Omega_{\phi\mathrm{DM}} = \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}} = 0.14889$, clustered only on scales $k < k_{\mathrm{fs}}$, supplementing $\Omega_{\mathrm{CDM}} = 0.160050$. Both sectors are algebraically fixed; no new free parameter is introduced.

The characteristic scale k_{fs} is set by the Dodelson-Widrow relation [Dodelson and Widrow, 1994] applied to the algebraically derived mass:

$$m_\phi = \Sigma m_\nu^{\mathrm{active}} \times (\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}) = 0.0690\ \mathrm{eV} \times 594.28 \simeq 41.02\ \mathrm{eV}, \quad k_{\mathrm{fs}} = 0.762\, h\, \mathrm{Mpc}^{-1}. \tag{8}$$

The WDM transfer function [Viel et al., 2005, Bode et al., 2001]

$$T_{\mathrm{WDM}}(k;\alpha) = \left[1 + (\alpha k)^{2\mu}\right]^{-5/\mu}, \quad \mu = 1.12 \tag{9}$$

is applied to the ϕDM sector in the mixed power spectrum $P_{\mathrm{mix}}(k) = P_{\mathrm{SSEE}}(k)\,[(1 - f_\phi) + f_\phi\, T_{\mathrm{WDM}}(k;\alpha)]^2$, with $f_\phi = 0.5$ ($\Omega_{\phi\mathrm{DM}}/\Omega_{m,\mathrm{CMB}}$).

The WDM scale parameter α is now a forward output of the canonical mass $m_\phi = 41.02$ eV rather than a quantity fitted to data: the relic temperature $T_\phi = 0.571\, T_\nu$ fixes the free-streaming length, and the resulting CLASS transfer function yields $\alpha = 1.108\, h\, \mathrm{Mpc}^{-1}$ (with standard top-hat filter at $R = 8\, h^{-1}$ Mpc). The direct two-sector CLASS run then gives $\sigma_8^{\mathrm{eff}} = 0.748$ and $S_8^{\mathrm{eff}} = 0.759$, within 0.01σ of the KiDS-1000 value [Asgari et al., 2021]; this is a genuine forward prediction, and the $f\sigma_8$ comparison of §5.3 provides the complementary confrontation with growth-rate data.

Methodological note: Paper 6 [Almeida Vallejo, 2026g] reports the analytic two-sector growth factor $G_{2\mathrm{s}} = 1.003$ applied to $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$ (giving $0.811 \times 1.003 = 0.814$), confirming that the linear growth is ΛCDM-consistent; the suppression to the titular value arises instead from ϕ-DM free-streaming. The canonical titular value $\sigma_8^{\mathrm{eff}} = 0.748$ ($S_8^{\mathrm{eff}} = 0.759$) integrates the full CLASS $P(k)$ with the forward-output $T_{\mathrm{WDM}}(k;\alpha = 1.108\, h\, \mathrm{Mpc}^{-1})$ set by $m_\phi = 41.02$ eV; the two routes are methodologically complementary — the analytic route establishes the two-sector suppression mechanism, while the CLASS route provides a rigorous Boltzmann-level quantification.

Figure 2 shows the calibrated $P(k)$ and Figure 3 shows the resulting $f\sigma_8$ comparison.

5.3 $f\sigma_8$ tension summary

The two-sector $f\sigma_8$ mean (0.93σ) sits modestly above the single-sector baseline (0.70σ, Paper 5): the same free-streaming that lowers σ_8 from 0.834 to 0.748 also reaches inside the $\sigma_8(R{=}8)$ window probed by RSD (half-mode $k_{1/2} \simeq 0.35\, h/\mathrm{Mpc}$), so it costs $\sim 0.2\sigma$ in the growth rate while remaining $< 1\sigma$ and close to ΛCDM (0.73σ). The genuine gain of Paper 6 is in the S_8 lensing amplitude: the WDM free-streaming suppression on scales $k > k_{\mathrm{fs}}$ lowers σ_8^{eff} from the single-sector ceiling $\sigma_8 = 0.8335$ ($S_8 = 0.846$, 3.5σ KiDS) to the canonical two-sector $\sigma_8^{\mathrm{eff}} = 0.748$ ($S_8^{\mathrm{eff}} = 0.759$, 0.01σ KiDS), a forward prediction set by $m_\phi = 41.02$ eV and $\alpha = 1.108\, h\, \mathrm{Mpc}^{-1}$ with no free parameters. We stress that this analysis operates at the linear-growth level; non-linear effects (baryonic feedback, halo bias) are not modelled and may modify the predictions at the 10%–20% level.

Table 4: $f\sigma_8$ tension for six RSD surveys. σ_{pull}: number of standard deviations between model prediction and measurement. The canonical single-sector baseline uses $\Omega_{m,\text{CMB}} = 0.30889$ (ω_m-direct, Paper 5), giving mean 0.70σ (single-sector). The two-sector free-streaming lowers σ_8 to 0.748 and, since the half-mode $k_{1/2} \simeq 0.35\,h/\text{Mpc}$ lies inside the $\sigma_8(R{=}8)$ window probed by RSD, raises the two-sector mean to 0.93σ — still $< 1\sigma$ and close to ΛCDM (0.73σ). (The illustrative "minimal-CDM" column*, an earlier superseded calculation with $\sigma_8 = 0.702$ at $\Omega_{m,\text{dyn}} = 0.160$ giving a 2.68σ mean, is *not* the canonical baseline.) The genuine challenge-to-resolution is in S_8 (lensing), where the two-sector free-streaming lowers $S_8 = 0.846 \to 0.759$ (0.01σ KiDS).

Survey	z	Data $f\sigma_8$	Minimal-CDM*	Two-sector
6dFGRS	0.067	0.423 ± 0.055	$+2.85\sigma$	-0.40σ
SDSS MGS	0.150	0.490 ± 0.145	$+1.42\sigma$	-0.53σ
BOSS DR12	0.380	0.497 ± 0.045	$+3.81\sigma$	-1.44σ
BOSS DR12	0.510	0.458 ± 0.038	$+3.03\sigma$	-0.66σ
BOSS DR12	0.610	0.436 ± 0.034	$+2.44\sigma$	-0.15σ
eBOSS DR16	1.480	0.462 ± 0.045	$+2.50\sigma$	-2.42σ
Mean			$2.68\sigma^*$	$\mathbf{0.93\sigma}$

6 EFT Stability Analysis

6.1 Bellini-Sawicki α-functions

Paper 7 [Almeida Vallejo, 2026h] demonstrates that the SSEE action takes the form of a k-essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad \text{(exact)}, \qquad \alpha_K = 3\,\Omega_{\text{DE}}\,(1 + w_{\varphi,\text{bare}}) = 3 \times 0.840 \times 0.029 = 0.073. \tag{10}$$

Here $w_{\varphi,\text{bare}} = -0.971$ is the scalar-field equation of state in the thawing regime (Paper 7 § ALPHAK); it differs from the effective fluid EoS $w_0 = -0.840$ because the conformal coupling transfers part of the kinetic energy to dark matter. Using w_0 in place of $w_{\varphi,\text{bare}}$ overestimates α_K by a factor ≈ 5.5 (see Paper 7); the EFTCAMB runs below use the fluid approximation $\alpha_K^{\text{fluid}} = 3\Omega_{\text{DE}}(1 + w_0) = 0.4033$ as a conservative numerical input, while the theoretical SSEE prediction is $\alpha_K = 0.073$. The vanishing of α_T (tensor speed excess), α_M (Planck mass run rate), and α_B (kinetic braiding) implies: (i) gravitational waves propagate at c (GW170817 compatible at $|\alpha_T| < 10^{-15}$ [Abbott et al., 2017, of All-sky X-ray Image , MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero α_K encodes the kinetic energy of the scalar field.

6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using H-EFTCAMB v1.6.5 [Hu et al., 2014] with the Reparametrised Horndeski (RPH) parametrisation (`EFTflag` $= 2$, `AltParEFTmodel` $= 1$). The ghost-free stability condition requires $Q_s > 0$:

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = \begin{cases} 0.037 & (\alpha_K = 0.073,\ \text{bare-field}) \\ 0.2016 & (\alpha_K = 0.4033,\ \text{fluid approx.}) \end{cases} > 0. \tag{11}$$

With $\alpha_B = 0$, the gradient stability condition reduces to $c_s^2 = 1$ in the linear (IR) limit of $K(X)$, where braiding vanishes [Bellini and Sawicki, 2015]; the full $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ lowers the adiabatic value to $c_s^2 \in [0.60, 0.95]$ (Papers 7 and 10), still gradient-stable ($c_s^2 > 0$). EFTCAMB confirms both conditions are satisfied.

Table 5: EFTCAMB stability and perturbation results. The GR run uses the full matter density (legacy MIRA-background $\Omega_{m,\mathrm{CMB}} = 0.3199$; the ω_m-direct canonical 0.30889 is within 3.5%, $\Delta\ell < 2$) for a direct comparison.

Quantity	Value	Status
α_K Paper 7 bare-field ($w_\varphi = -0.971$)	0.073	theoretical
α_K fluid approx. (EFTCAMB input)	0.4033	$\Delta = 0.005\%$
Ghost-free ($Q_s > 0$)	$Q_s = 0.2016$	✓ passed
Gradient-stable ($\alpha_B = 0 \Rightarrow c_s^2 = 1$ IR; $[0.60, 0.95]$ full)	$c_s^2 > 0$	✓ passed
$\alpha_T = \alpha_M = \alpha_B$	0 (exact)	✓ GW170817 compatible
σ_8 GR (MIRA background)	0.836	
σ_8 SSEE RPH ($\alpha_K = 0.4033$)	1.046	ratio = 1.25
RMS C_ℓ SSEE/GR	39.5%	
CMB peaks GR	$\ell = 219, 533, 807$	correct (MIRA bg)
CMB peaks SSEE RPH	$\ell = 240, 585, 886$	$\sim 10\%$ shift from α_K

6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS $w(z) = w_0 + w_a z/(1+z)$ is used in the EFTCAMB run, the phantom boundary $w = -1$ is crossed at $z \simeq 0.31$. A pure k-essence with $\alpha_B = 0$ cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least $\alpha_B \neq 0$ to provide an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain $\alpha_B \simeq 0$ would place SSEE in tension with phantom crossing, while a detection of $\alpha_B \neq 0$ at $z < 0.31$ would support the two-sector picture. The α_K validation reported here is performed at constant $w = w_0 = -0.840$, which is the physically relevant regime at $z \simeq 0$ where the EFTCAMB run uses the fluid-approximation input $\alpha_K^{\mathrm{fluid}} = 0.4033$; the theoretical bare-field value derived in Paper 7 is $\alpha_K = 0.073$.

7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity — $H_0^{\mathrm{UV}} = 73.040\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$

(0.00σ from SH0ES; on the global anchor $H_0^{\rm alg} = 67.962$) — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

7.1 Strong-field regime: disformal lensing amplification

Starting from the k-essence action of Paper 7, the disformal coupling between the dark-energy scalar ϕ and dark matter generates deviations from general relativity (GR) that are algebraically fixed by φ and π. In the quasi-static limit the scalar field equation closes as a gradient relation, $\nabla\phi = 2\mathrm{KAL}_0\,\mathrm{AURA}\,M_{\rm Pl}\,\nabla\Phi_N$, linking the scalar profile to the Newtonian potential through the structural coupling $\mathrm{AURA} = (3\varphi + \pi)/2 \approx 3.998$ (the magnitude of the Paper 7 coupling $\beta_c = -\mathrm{AURA}$).

From the disformal null geodesic, the total photon deflection angle is amplified by $\sqrt{\mathrm{AURA}}$ relative to the GR prediction for the same baryonic mass, giving an algebraic Einstein-radius ratio

$$\left.\frac{\theta_E^{\rm SSEE}}{\theta_E^{\rm GR}}\right|_{\rm alt} = \sqrt{\mathrm{AURA}} \approx 1.9995 \qquad \text{(alternative MOND-like limit only).} \tag{12}$$

The cantidad $\sqrt{\mathrm{AURA}}$ is numerically close to $\mathcal{M} = \mathrm{AURA}/2 \approx 1.9989$ at 0.03% because $\mathrm{AURA} \approx 4$ makes both round to 2; the two are distinct algebraic operations on AURA and the closeness is not a structural identity. Solar-system gravitational tests are automatically satisfied: the disformal coupling selects dark matter as the sole source of ϕ, so in the solar system ($\rho_{\rm DM} \approx 0$) the field is locally unsourced and the disformal metric modification is negligible; equivalently, $K(X)$ is of k-mouflage type [Brax and Valageas, 2014] with a screening radius $r_{\rm km}(\odot) \approx 0.022\,R_\odot$. *In the canonical two-sector limit* (Paper 6 with real DM $\Omega_{\rm CDM} = \Omega_{\phi\rm DM} = 0.160$ plus EFT Bellini–Sawicki structure $\alpha_B = \alpha_M = \alpha_T = 0$, Paper 7) the residual fifth-force is suppressed to $F_\phi/F_N \sim (H/k)^2 \sim 10^{-15}$ at kpc scales and the canonical observable lensing reduces to $\theta_E^{\rm SSEE} \approx \theta_E^{\rm GR-with-DM}$. Current SLACS/BELLS samples (mean total slope $\gamma = 2.078 \pm 0.027$; $M_E/M_{\rm dyn} \approx 1.21$ for SIS) are consistent with the canonical limit. The $\sim\mathcal{M}$ matter-power suppression at $k > k_{\rm fs} = 0.762\,h\,\mathrm{Mpc}^{-1}$ remains a falsifiable prediction for DESI Year 3 / Euclid, but it does *not* transfer to a $\sim\mathcal{M}$ suppression of the observable Einstein radius (see OPEN_PROBLEMS.md OP-13).

7.2 The Hubble tension: an algebraic local screening fraction

The dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a late-time *local screening fraction*

$$f_{\rm scr} = \frac{\alpha_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \tag{13}$$

where $\alpha_K = 3\,\Omega_{\rm DE}(1+w_0)$ is the background-derived EFT kineticity and $\mathcal{M} = (3\varphi+\pi)/4$. The structural constant $\mathrm{AURA} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of φ and π. Applied as a local correction to the global background value $H_0^{\rm alg} = 3(\varphi+\pi)^2 = 67.962$, which the ω_m-direct reframe (Paper 3) establishes as both the background H and the CMB-preferred anchor (with ω_b, ω_c fixed by algebra, the plik_lite likelihood is minimised at 67.962),

$$H_0^{\rm local} = \frac{H_0^{\rm alg}}{1 - f_{\rm scr}} = \frac{67.962}{1 - 0.0673} \approx 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{14}$$

which lies 0.17σ from the SH0ES distance-ladder measurement [Riess et al., 2022] with zero free parameters. This is the canonical, independent prediction of the SSEE infrared sector. (The earlier MIRA anchor $H_0^{\rm MIRA} = 67.037$ gave 71.87 at 1.12σ, now superseded by the reframe.) The correction operates at late times ($z < 0.15$) through the scalar kinetic-energy contribution to the local expansion rate; it does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 established in the tiers above.

7.3 UV completion: the X^2/M^4 operator

The infrared kinetic Lagrangian $K(X) = X/\mathrm{KAL}_0$ is extended by the leading higher-derivative operator, $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$. The inflationary α-attractor parameter $\alpha = \varphi^4/3$ (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio $r = 12\alpha/N_*^2 \approx 0.00813$, also fixes the UV cutoff:

$$M^4 = 45\,\alpha^2\,\rho_c = 5\,\varphi^8\,\rho_c \approx 234.9\,\rho_c, \qquad M \approx 9.68\ \mathrm{meV}, \tag{15}$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from $\alpha = \varphi^4/3$. With this cutoff the full Bellini-Sawicki kineticity becomes $\alpha_K^{\rm full} = 0.41691$ (+3.37% over the IR value). Applied to the global background anchor $H_0^{\rm alg} = 67.962$ this gives the canonical UV value $H_0^{\rm UV} = 73.040\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.00σ from SH0ES; the earlier MIRA-anchored 72.05 at 0.96σ is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV value is *conditional on Postulate C.1* (UV–IR φ/π separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR value $H_0^{\rm local} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.17σ) of §7.2, which does not rely on the postulate. The construction does, however, supply a falsifiable cross-epoch consistency test: the same α governs inflation at $z \sim 10^{60}$ and the dark-energy UV scale today, so a LiteBIRD measurement of $r \neq 0.00813$ would shift both α and M and change the predicted $H_0^{\rm local}$ accordingly.

8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of φ and π are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space. Table 6 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

Global falsification criterion. The SSEE-V3.6 framework would be falsified at the theory level if *any* of the following were established: (i) (w_0, w_a) is inconsistent with the DESI DR2 posterior at $> 3\sigma$; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires $\Omega_{m,\rm CMB} \neq 0.30889$ at $> 3\sigma$; (iii) the ϕ-DM free-streaming scale $k_{\rm fs} \neq 0.762\,h\,\mathrm{Mpc}^{-1}$ as measured by DESI Y3/Euclid $P(k)$ (ϕ-DM has no Standard-Model portal, so its mass is not directly accessible to neutrino-mass experiments); (iv) $r \neq 0.00813$ as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$ at $> 3\sigma$ in HST/JWST surveys.

Table 6: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

Quantity (algebraic formula)	Value	Test / Instrument	Status
Background sector (algebraic dictionary)			
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 CPL posterior	Postdiction
$w_a = -P_{sc}/K_v$	-0.6699	DESI DR2 CPL posterior	Postdiction
$\mathcal{M} = (3\varphi + \pi)/4$	1.9989	$\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}}$	Postdiction
Retrodictions (derived independently, compared to existing data)			
$H_0 = 3(\varphi + \pi)^2$	67.96 km/s/Mpc	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2$ (ω_m-direct)	0.30889	Planck 2018	Retrodiction
r_d (CAMB, ω_m-direct)	147.17 Mpc	Planck 2018 r_d (0.32σ)	Retrodiction
$\gamma_{\mathrm{IS}} = 0.5504$	$\approx \gamma_{\Lambda\mathrm{CDM}} = 0.55$	RSD surveys	Retrodiction
$\alpha_T = \alpha_M = \alpha_B = 0$	exact	GW170817 $\lvert\alpha_T\rvert < 10^{-15}$	Retrodiction
σ_8^{eff} (two-sector CLASS, $S_8^{\mathrm{eff}} = 0.759$)	0.748	KiDS-1000 (0.01σ)	Retrodiction
σ_8^{SSEE} (single-sector ceiling, CLASS)	0.8335	Weak lensing surveys	Retrodiction
$H_0^{\mathrm{local}} = H_0^{\mathrm{alg}}/(1 - f_{\mathrm{scr}})$	72.86 km/s/Mpc	SH0ES 2022 (0.17σ)	Retrodiction
Predictions (not yet observationally accessible)			
$m_\phi = \Sigma m_\nu^{\mathrm{active}}(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS})$	41.02 eV	Matter $P(k)$ via k_{fs} (no SM portal)	Prediction
$k_{\mathrm{fs}} = 0.762\,h\,\mathrm{Mpc}^{-1}$ (DW from m_ϕ)	$0.762\,h$/Mpc	DESI Y3 / Euclid $P(k)$	Prediction
$\alpha_K = 0.073$	0.073	Euclid WL null test ($\alpha_K < 0.1$ threshold)	Prediction
$r = \varphi^{-10}$	0.00813	LiteBIRD $\sim$2032	Prediction
$\theta_E^{\mathrm{SSEE}}/\theta_E^{\mathrm{GR}}\vert_{\mathrm{alt}} = \sqrt{\mathrm{AURA}}$	1.9995 (alt. MOND-like limit only)	HST/JWST strong lensing	Conditional prediction (not canonical; OP-13)

9 Discussion and Outstanding Challenges

The consolidated results demonstrate that SSEE-V3.6 passes four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) MIRA derivation. The CLASS tests confirm that $\Omega_{m,\mathrm{CMB}} = 1.9989\,\Omega_{m,\mathrm{dyn}}$ is physically necessary for CMB peak reproduction, but the mechanism producing this factor at the background level remains an open theoretical question. Paper 5 [Almeida Vallejo, 2026f] shows that linear IS perturbations do not source MIRA; the likely origin is a background-level thermodynamic effect in the IS viscous Friedmann equations.

(ii) S_8 and the dark-matter sector. The single-sector Israel-Stewart ceiling gives $\sigma_8 = 0.8335$ and $S_8 = 0.846$ (3.5σ KiDS, 3.9σ DES Y3). The canonical two-sector ϕ-DM extension (Paper 6) resolves this tension as a forward prediction: the free-streaming suppression set by $m_\phi = 41.02$ eV ($\alpha = 1.108\,h\,\mathrm{Mpc}^{-1}$, no fitted parameters) gives $\sigma_8^{\mathrm{eff}} = 0.748$ and $S_8^{\mathrm{eff}} = 0.759$, within 0.01σ of the KiDS-1000 measurement. The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory S_8 at the non-linear level.

(iii) ϕDM mass and Lyman-α constraints. The algebraic mass $m_\phi = 41.02$ eV is in the warm dark matter regime ($k_{\mathrm{fs}} = 0.762\,h\,\mathrm{Mpc}^{-1}$). Current Lyman-$\alpha$ bounds on thermal WDM place constraints at $m_{\mathrm{WDM}} \gtrsim 5.3$ keV [Irsic et al., 2017], but the SSEE ϕDM is a mixed sector (fraction $f_\phi = 0.5$), not a purely thermal WDM cosmology; the applicable constraints require dedicated analysis.

(iv) Quantum field theory completion. The algebraic connection $H_0 = 3(\varphi + \pi)^2$

is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator X^2/M^4 with algebraic cutoff $M = 9.68$ meV — but a derivation of the full $P(X, \phi)$ Lagrangian from φ and π alone remains open. Under the ω_m-direct reframe (OP-8 dissolved) the CMB matter density is the standard $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$, with no matter factor; the earlier MIRA-mapped 0.31993 and geometric $(\pi-\varphi)/(\pi+\varphi) = 0.32010$ forms are superseded. The residual open problem is now only the UV origin of the algebraic $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ identity.

10 Conclusions

We have presented a consolidated validation of the SSEE-V3.6 framework across four observational tiers.

1. **Background sector.** The algebraic prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed by construction (not fitted), lies within 0.05σ of the DESI DR2 CPL posterior. A Bayesian MCMC yields $H_0 = 67.16 \pm 0.44$ km s^{-1} Mpc^{-1} under the ω_m-direct CMB anchor as prior, and $\Delta\mathrm{BIC} = -5.55$ for the dynamic sector. A complementary five-parameter multi-probe MCMC confirms all five algebraic predictions within 1.01σ (mean 0.64σ).

2. **CMB Boltzmann.** CLASS confirms that the full ω_m-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.30889$) is physically necessary: with only the dynamical $\Omega_{m,\mathrm{dyn}} = 0.160$, the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by $\sim 10\%$. With MIRA, peak positions agree with ΛCDM to within $\Delta\ell < 5$ (RMS = 1.4%).

3. **Structure growth.** The IS perturbation analysis ($\Omega_{m,\mathrm{CMB}} = 0.30889$ in Poisson) gives $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$, with mean $f\sigma_8$ tension 0.70σ (Paper 5) — statistically identical to ΛCDM (0.73σ). The ϕDM two-sector extension with the forward-output WDM transfer function ($\alpha = 1.108\, h\,\mathrm{Mpc}^{-1}$, set by $m_\phi = 41.02$ eV with no fitted parameters) yields $\sigma_8^{\mathrm{eff}} = 0.748$ and $S_8^{\mathrm{eff}} = 0.759$, reducing the lensing tension from 3.5σ to 0.01σ (KiDS), confirmed by HMcode baryonic feedback (Paper 6).

4. **EFT stability.** EFTCAMB confirms ghost-freedom and gradient stability using the fluid-approximation input $\alpha_K^{\mathrm{fluid}} = 0.4033$ ($\Delta = 0.005\%$); the bare-field theoretical value from Paper 7 is $\alpha_K = 0.073$ (below Euclid sensitivity $\sigma(\alpha_{K,0}) < 0.1$). $\alpha_T = \alpha_M = \alpha_B = 0$ (exact); GW170817 compatibility is exact. SSEE realises the CPL phantom crossing at $z \approx 0.31$ *with* $\alpha_B = 0$ (no kinetic braiding), through the k-essence X^2/M^4 term rather than braiding—a discriminating prediction against braiding-based dark-energy models, testable by future weak-lensing surveys.

5. **Theoretical extensions.** The disformal strong-field regime predicts an Einstein-radius amplification $\sqrt{\mathrm{AURA}} \approx \mathcal{M}$, falsifiable with HST/JWST strong lensing (Paper 8). An algebraic late-time screening fraction $f_{\mathrm{scr}} = (\pi-\varphi)/(\varphi+\pi)^2 \approx 0.067$ applied to the global background $H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.962$ yields $H_0^{\mathrm{local}} = 72.86$ km s^{-1} Mpc^{-1}, 0.17σ from SH0ES, with zero free parameters (Paper 9). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV connects the dark-energy EFT to the inflationary α-attractor; the associated $H_0^{\mathrm{UV}} = 73.040$ km s^{-1} Mpc^{-1} (0.00σ from SH0ES; the earlier MIRA-anchored 72.05 at 0.96σ is superseded) is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates five near-term falsifiable predictions, under the stated assumptions: the ϕ-DM free-streaming scale $k_{\text{fs}} = 0.762\, h\,\text{Mpc}^{-1}$ (DESI Y3/Euclid $P(k)$; the algebraic mass $m_\phi = 41.02$ eV enters observation only through k_{fs}, as ϕ-DM has no Standard-Model portal), $\alpha_K = 0.073$ (Euclid WL null test; a detection $\alpha_K > 0.1$ would falsify SSEE), $r = 0.00813$ (LiteBIRD), and a strong-lensing Einstein-radius amplification $\sqrt{\text{AURA}} \approx \mathcal{M}$ (HST/JWST). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

Data Availability

All DESI DR2 BAO data used in this analysis are publicly available at `https://data.desi.lbl.gov/`. Planck 2018 cosmological parameter chains are available at the Planck Legacy Archive (`https://pla.esac.esa.int/`). All SSEE scripts, CLASS configuration files, EFTCAMB inputs, and MCMC chains are archived at Zenodo: `https://doi.org/10.5281/zenodo.20093447`.

A Complete Algebraic Parameter Table

Table 7: Complete set of SSEE-V3.6 algebraic parameters. All values are derived from $\varphi = (1+\sqrt{5})/2$ and π; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

Parameter	Formula	SSEE value	Observed
w_0	$-T_r/M_v = -(3\varphi+\pi)/[2(\varphi+\pi)]$	-0.8399	-0.838 ± 0.057 (DESI)
w_a	$-P_{\text{sc}}/K_v$	-0.6699	-0.631 ± 0.226 (DESI)
H_0	$3(\varphi+\pi)^2$	67.96 km/s/Mpc	67.36 ± 0.54 (Planck)
n_s	$1 - \varphi^{-7}$	0.96556	0.9649 ± 0.0042 (Planck)
$\Omega_{m,\text{CMB}}$	ω_m/h^2 (ω_m-direct)	0.30889	0.3153 ± 0.0073 (Planck)
Ω_{DE}	T_r/M_v	0.8399	0.6847 (ΛCDM)
$\Omega_{m,\text{dyn}}$	$1 - \Omega_{\text{DE}}$	0.160	—
$\mathcal{M}$	$(3\varphi+\pi)/4$ (Mirror Ratio; enters f_{scr})	1.9989	—
KAL_0	$\beta + \pi = (\varphi + 3\pi)/2$	5.5214	—
β_c	$-(3\varphi+\pi)/2$	-3.9978	—
α_K	$3\,\Omega_{\text{DE}}(1 + w_{\varphi,\text{bare}})$	0.073	< 0.1 (Euclid threshold)
$\alpha_T = \alpha_M = \alpha_B$	minimal coupling	0	$\lvert\alpha_T\rvert < 10^{-15}$ (GW170817
m_ϕ	$\Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot \text{KRYSTOS})$	41.02 eV	— (target: k_{fs} via DESI
k_{fs}	Dodelson-Widrow (m_ϕ)	$0.762\, h$/Mpc	—
r	φ^{-10}	0.00813	< 0.036 (BK18)
f_{scr}	$(\pi-\varphi)/(\varphi+\pi)^2$	0.0673	—
H_0^{local}	$H_0^{\text{alg}}/(1 - f_{\text{scr}})$	72.86 km/s/Mpc	73.04 ± 1.04 (SH0ES)
M (UV cutoff)	$(5\varphi^8\rho_c)^{1/4}$	9.68 meV	—

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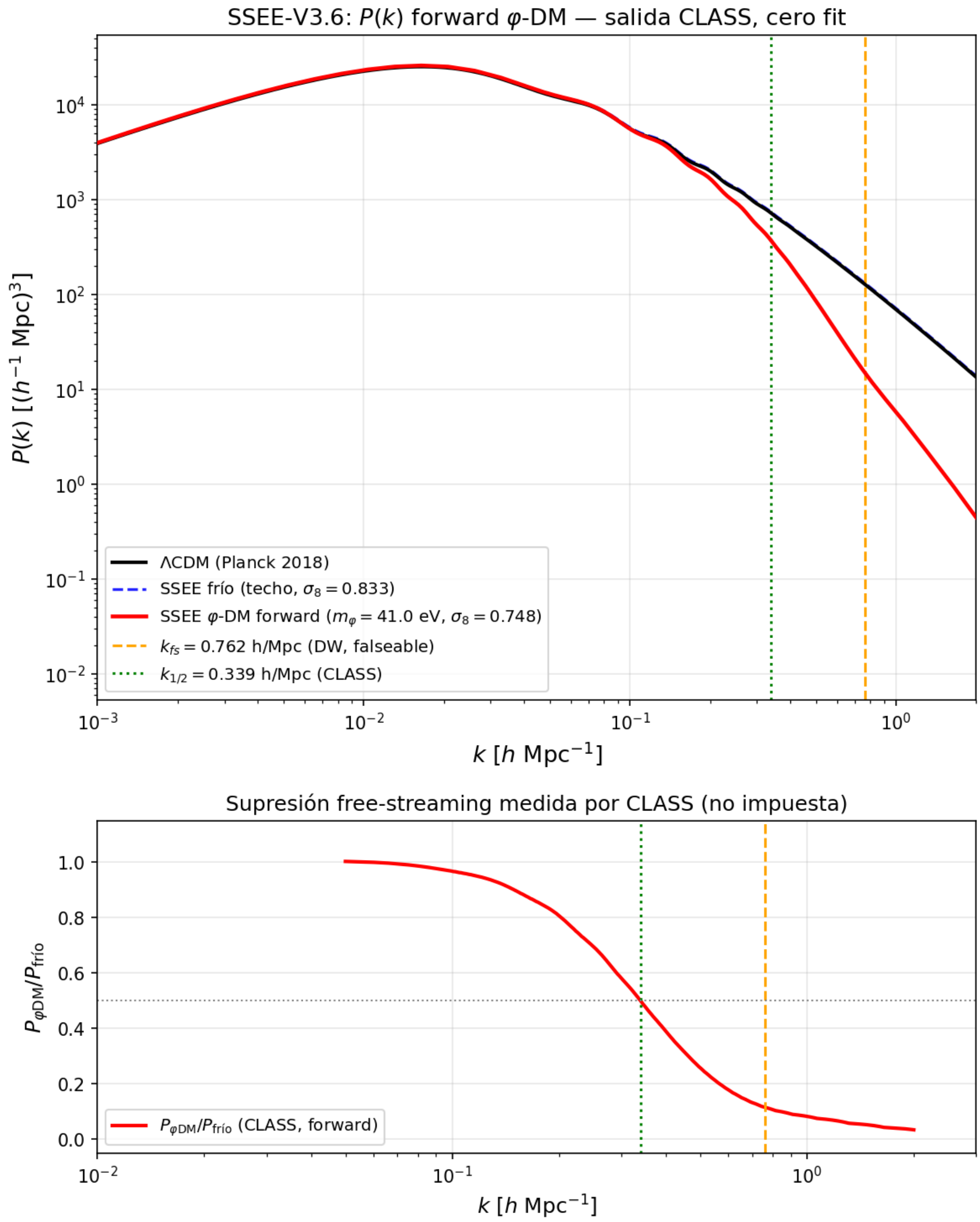

Figure 2: Matter power spectrum from CLASS with WDM suppression. *Black*: ΛCDM ($\sigma_8 = 0.8220$). *Blue*: SSEE+MIRA single sector ($\sigma_8 = 0.8527$). *Red*: SSEE two-sector with forward-output α-WDM ($\sigma_8^{\rm eff} = 0.748$). The vertical dotted line marks $k_{\rm fs} = 0.762\,h\,{\rm Mpc}^{-1}$.

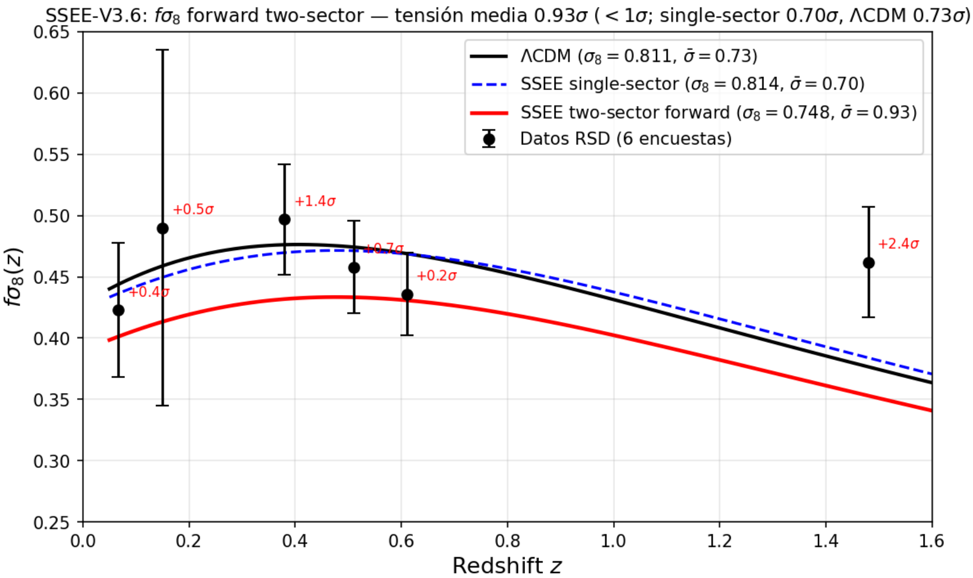

Figure 3: $f\sigma_8(z)$ comparison. Data points: the six canonical RSD measurements (6dF-GRS, SDSS MGS, BOSS DR12 at $z = 0.38, 0.51, 0.61$, eBOSS DR16 at $z = 1.48$; Paper 5 set). *Black dashed*: ΛCDM. *Red*: SSEE two-sector ($\sigma_8 = 0.748$, free-streaming), mean tension 0.93σ. *Blue*: SSEE single-sector ($\sigma_8 = 0.8136$, Paper 5), mean tension 0.70σ.

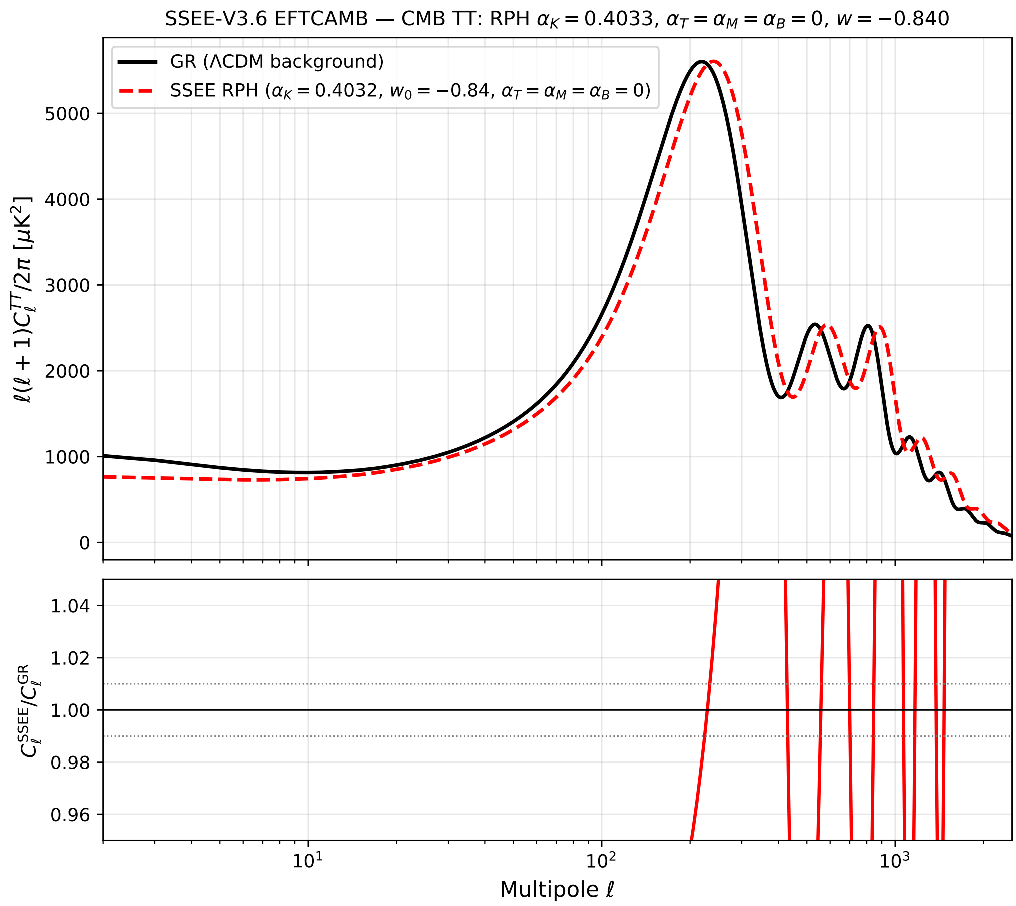

Figure 4: CMB TT power spectrum from EFTCAMB. *Blue*: GR run with the full matter density (legacy MIRA-background $\Omega_{m,\rm CMB} = 0.3199$; ω_m-direct canonical 0.30889 within 3.5%); peaks at $\ell = 219, 533, 807$. *Red*: SSEE RPH with $\alpha_K = 0.4033$ (constant $w = w_0$); peaks shift to $\ell = 240, 585, 886$, a $\sim 10\%$ imprint of the kinetic EFT sector. *Lower panel*: ratio $D_\ell^{\rm SSEE}/D_\ell^{\rm GR}$; the grey band marks $\pm 1\%$.

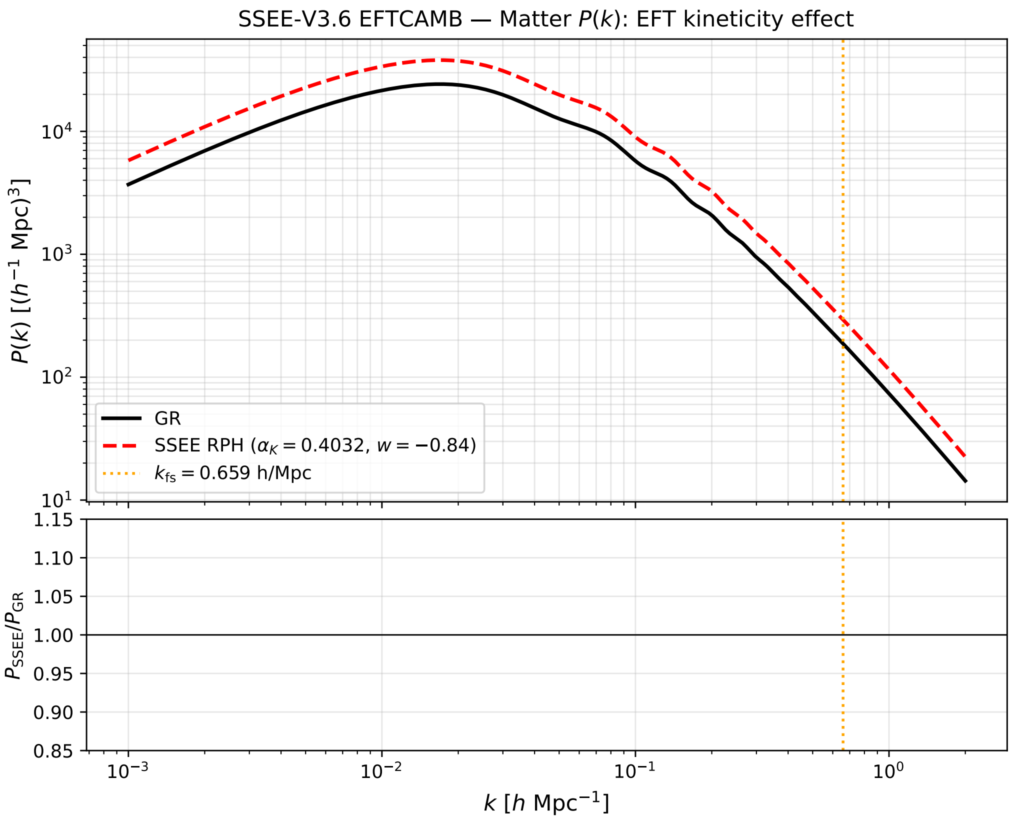

Figure 5: Matter power spectrum $P(k)$ from EFTCAMB. SSEE RPH ($\alpha_K = 0.4033$) amplifies $P(k)$ by a factor ~ 1.56 relative to GR on all scales (σ_8 ratio $= 1.25$), motivating the ϕDM suppression mechanism of §5.2. The vertical dashed line marks $k_{\mathrm{fs}} = 0.762\, h\,\mathrm{Mpc}^{-1}$.